

5. On detonation, the explosive RDX produces carbon monoxide, steam and nitrogen.



RDX
 M_r 222

Calculate the mass of RDX that would give 1.00 m^3 of gaseous products measured at 100°C and at 1 atm. Give your answer to an **appropriate** number of significant figures. [2]

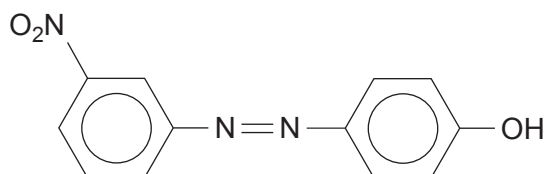
(1 mol of any gas occupies a volume of 30.6 dm^3 under these conditions)

Mass of RDX = g

10



5. The formula of an azo dye, produced by reacting a diazonium compound with phenol, is shown below.



Write the formula of the amine that is used to produce this diazonium compound.

[1]

6. Ethanal reacts with hydrogen cyanide to give 2-hydroxypropanenitrile, $\text{CH}_3\text{CH}(\text{OH})(\text{CN})$.

(a) State the type of mechanism occurring during this reaction.

[1]

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(b) Draw the structures of the two enantiomers of 2-hydroxypropanenitrile.

[1]

